ECE 340: Semiconductor Devices

Lecture 5: Energy Bands in Solids

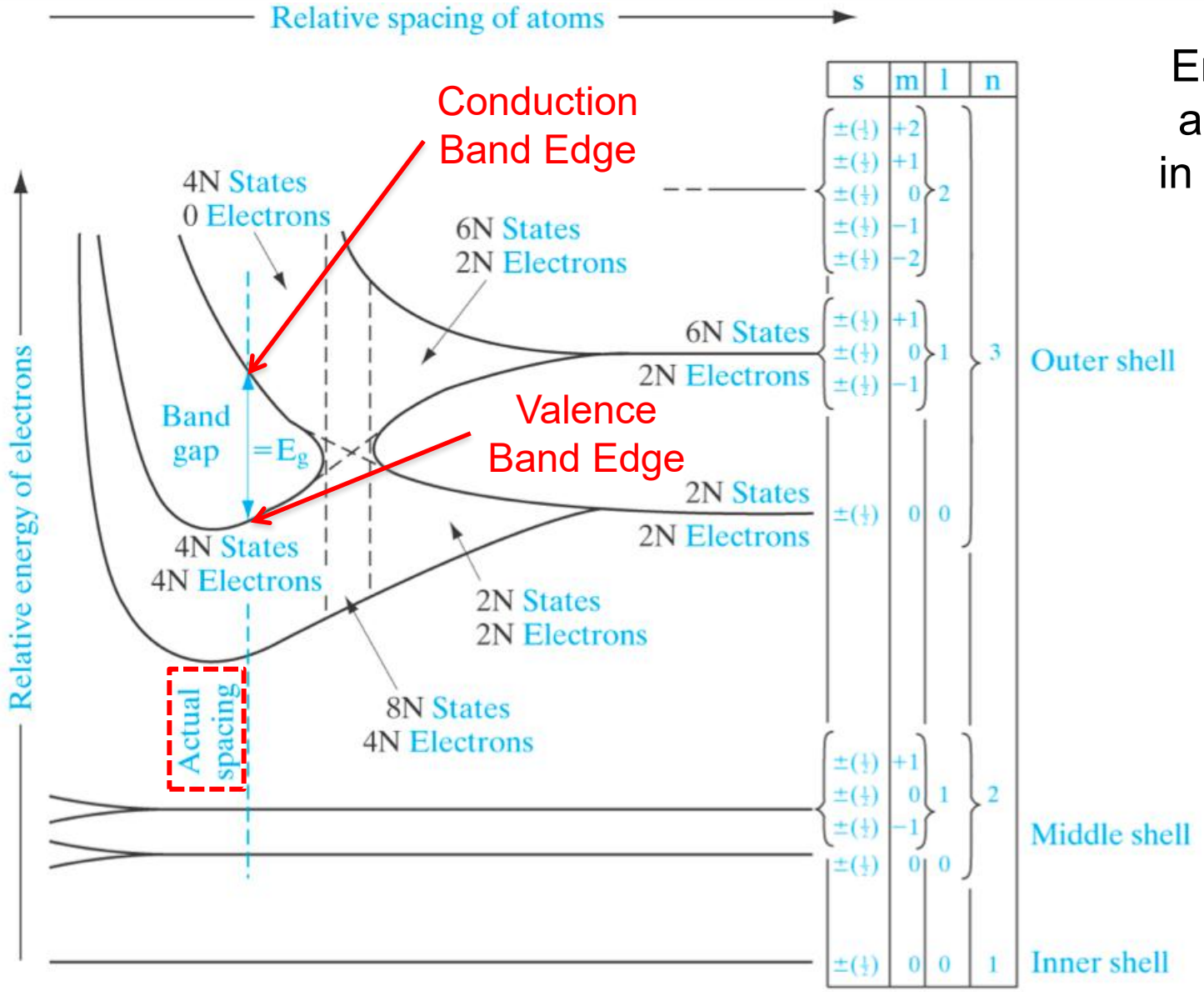
Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

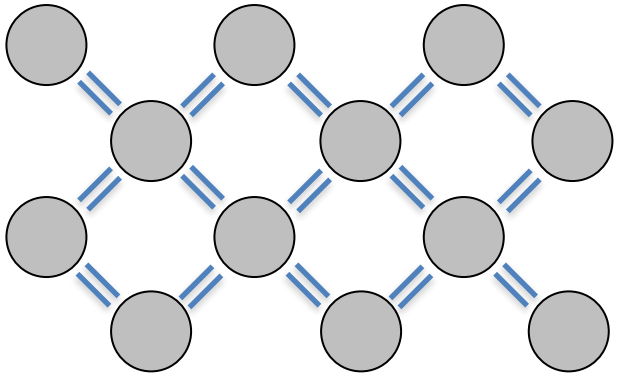
Office Hours: Thursdays 12p-1p, 2256 HMNTL

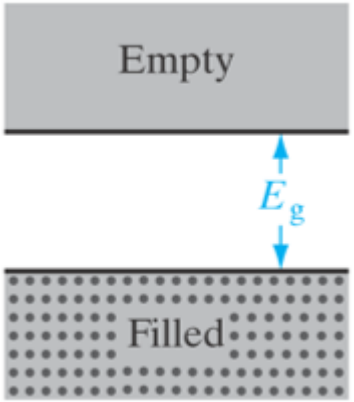
Recap: Formation of energy bands in Si



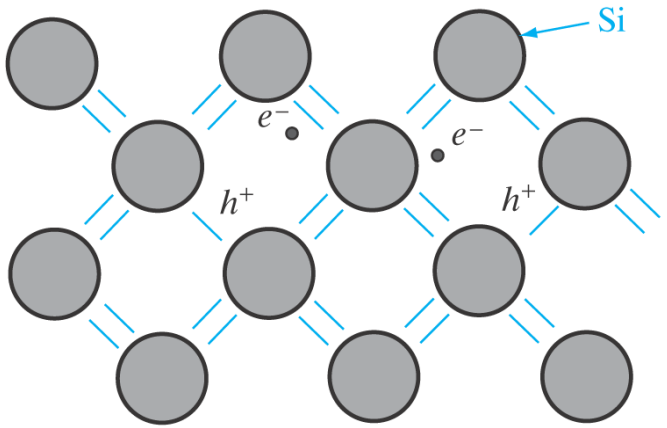
Energy bands and bandgap in silicon lattice

Recap: Electrons and Holes in "Perfect" Semiconductors





$n = p = n_i$



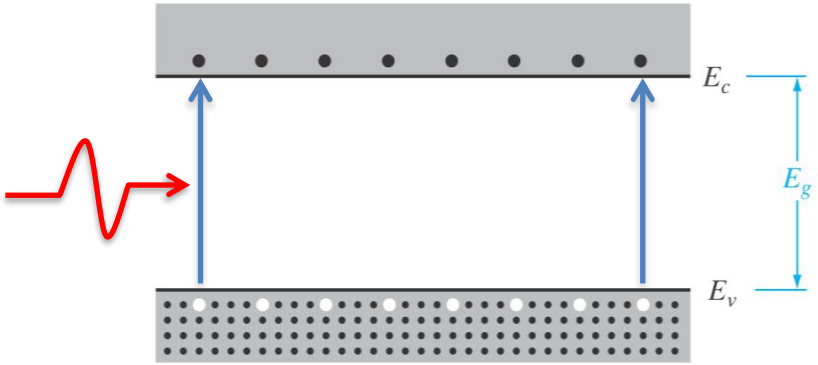


Figure 3.7

Electron–hole pairs in a semiconductor.

Effective Mass

Electron energy-momentum in free space

- Particle's momentum ***p***

$$p=mv \rightarrow v=p/m$$

- Kinetic energy ***E*** of a particle

$$E = \frac{mv^2}{2} = \frac{1}{2}\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

- De Broglie relation

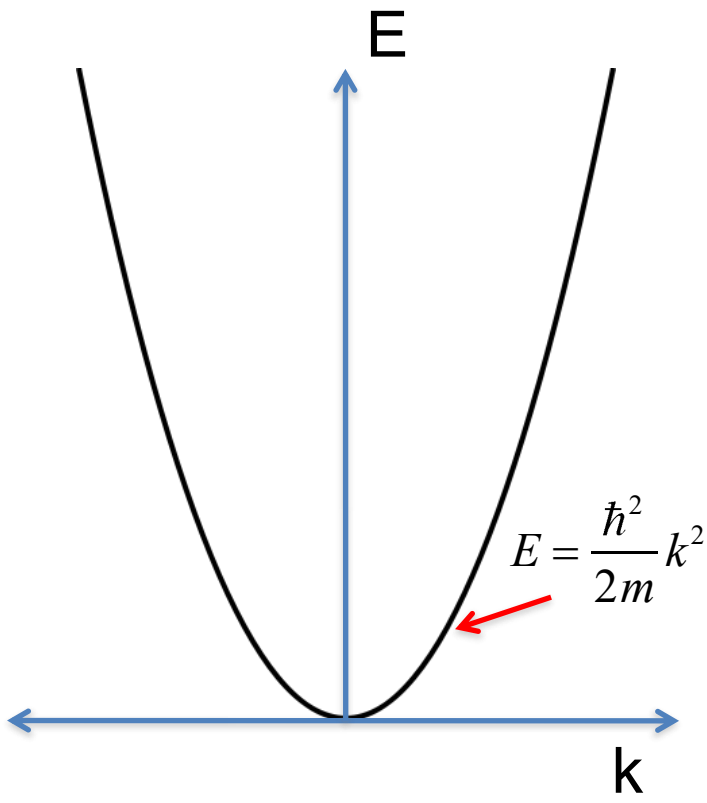
$$\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}$$

- Define wavenumber ***k***

$$k = \frac{2\pi}{\lambda} \quad \text{, then } p = \hbar k$$

$$E = \frac{p^2}{2m} = \frac{\hbar k^2}{2m}$$

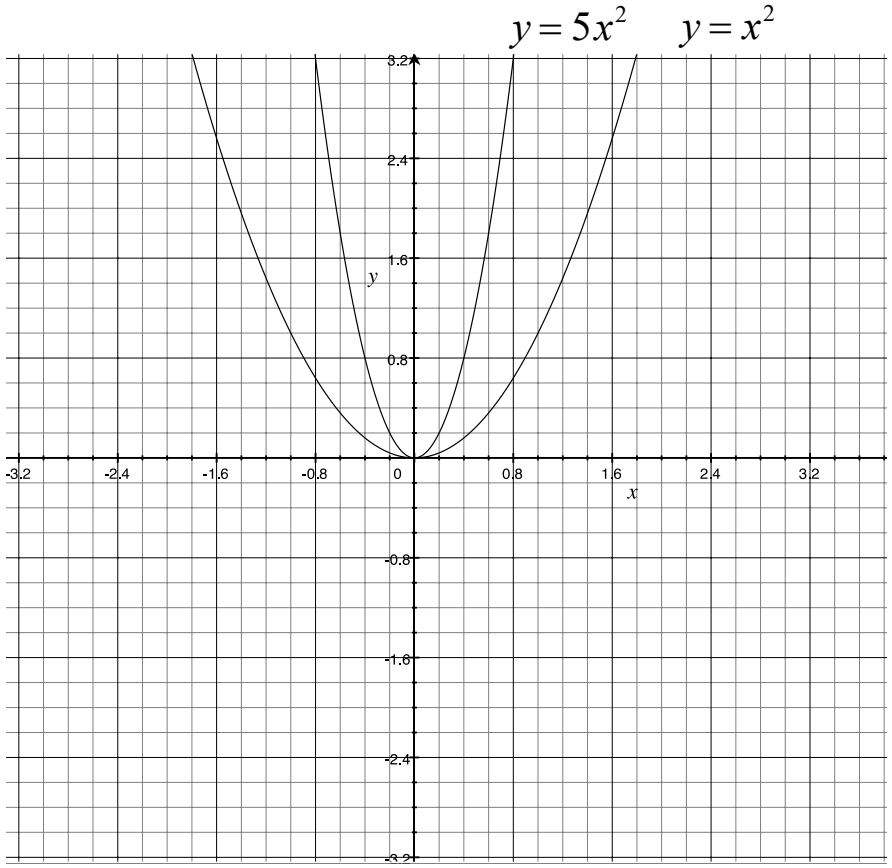
Classical Mechanics: $E = \frac{p^2}{2m}$



What would this look like if the mass were smaller? or negative?

Equation for a Parabola

$y = ax^2$ □ Curvature at origin related to "a" coefficient



How would larger ***m*** be reflected in the *E*-*k* relationship ?

$$E = \frac{\hbar^2}{2m} k^2$$

What if ***m*** is negative?

Electron in crystal lattice

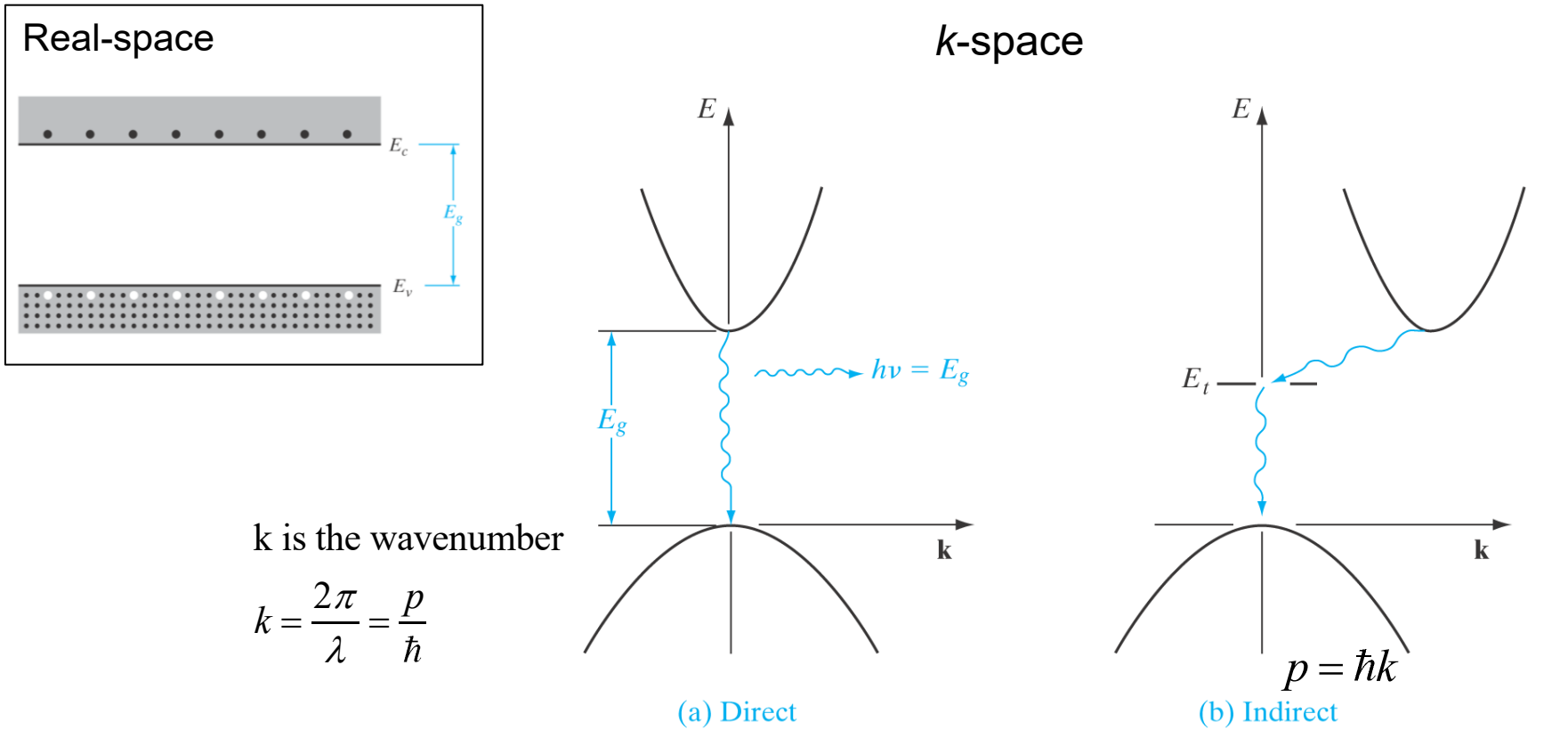


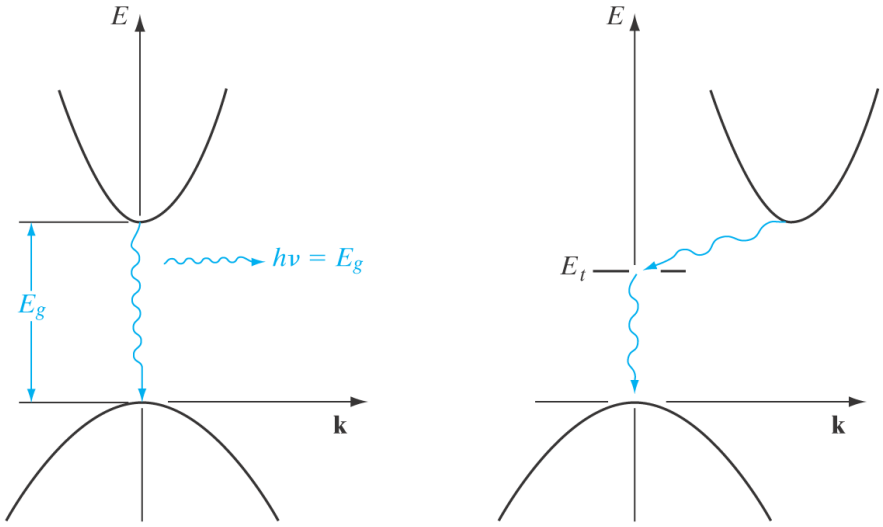
Figure 3.5

Direct and indirect electron transitions in semiconductors: (a) direct transition with accompanying photon emission; (b) indirect transition via a defect level.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Electron and Hole Effective Mass

- In a semiconductor, the E(k) relationship looks parabolic near the conduction band minima and the valence band maxima
- **The curvature of the parabola differs from the free space curvature**, and depends upon which minima the electron or hole is in
- Near a minima, the E(k) relationship is: $$E = \frac{\hbar^2}{2m^*}\mathbf{k}^2$$
- This is similar to the E(k) relationship for a free electron with a mass m* instead of m_o
- The parameter **m*** is the **effective mass**
- What about valence band? → **Positive "Hole" effective mass**



(a) Direct (b) Indirect

Si and GaAs in Momentum (k-) Space

For a given k_x, k_y, and k_z the electron has some energy E(eV)

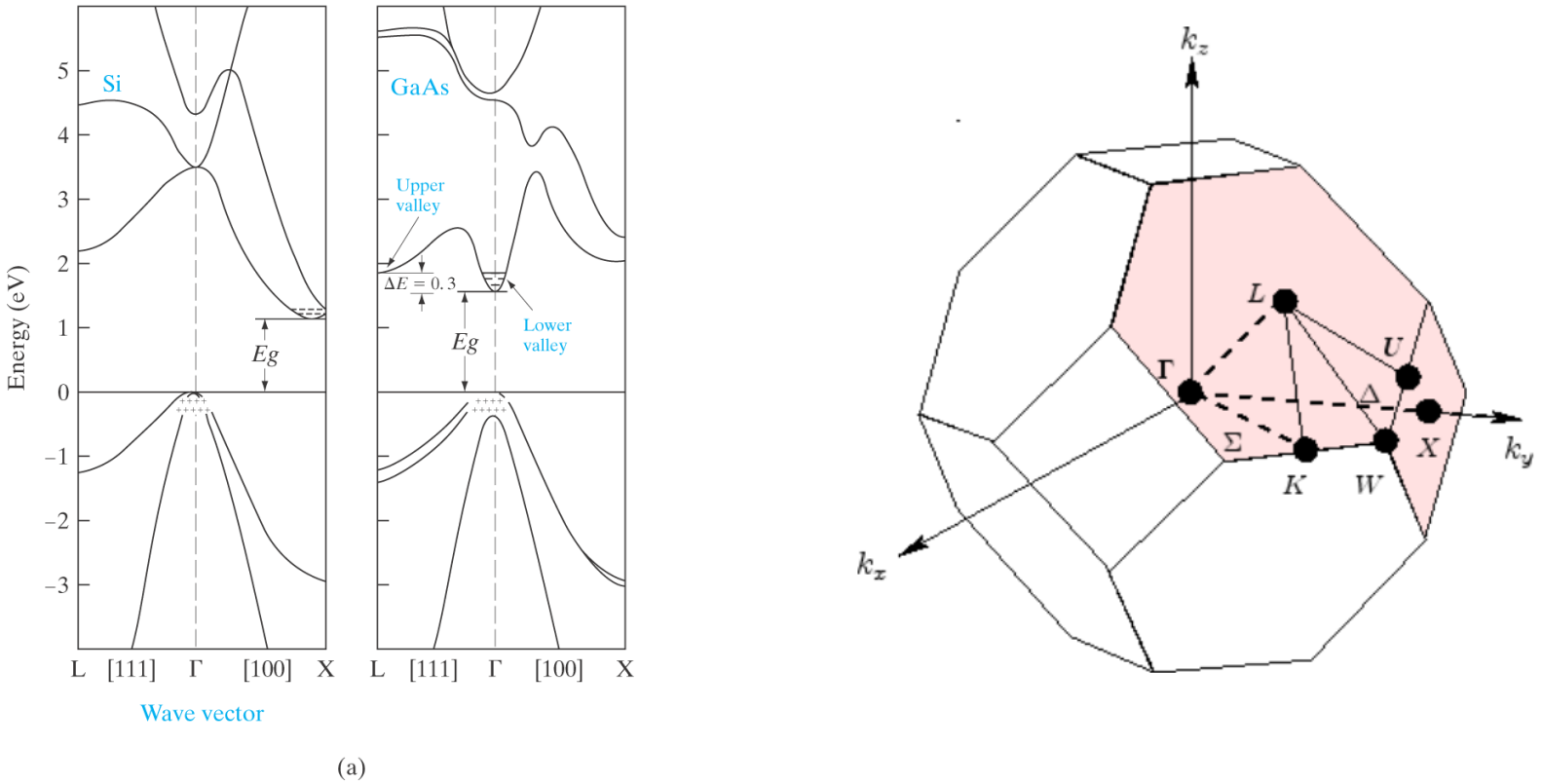


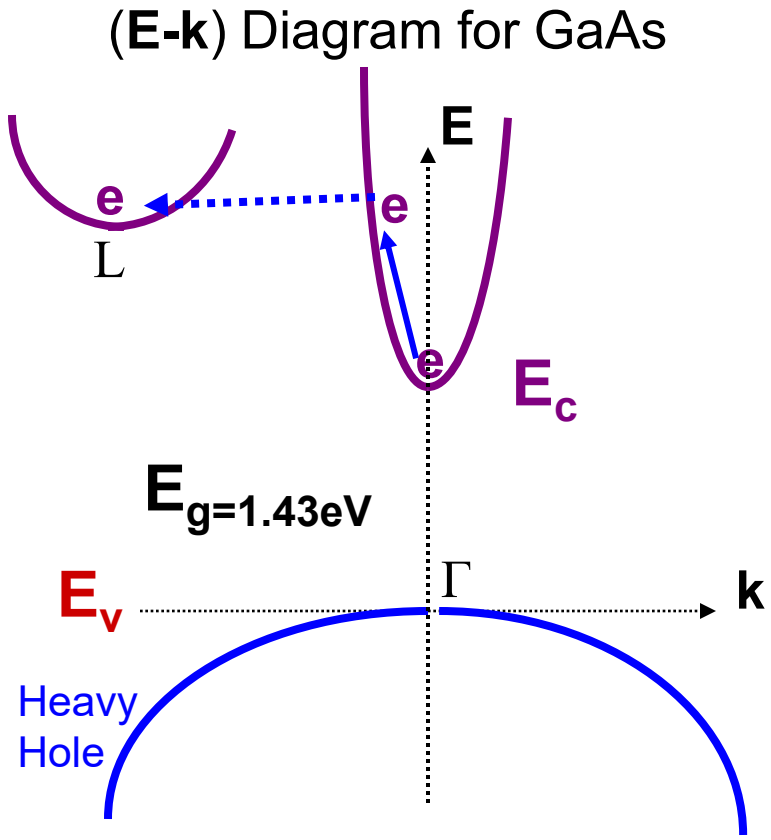
Figure 3.10

Realistic band structures in semiconductors: (a) Conduction and valence bands in Si and GaAs along [111] and [100]; (b) ellipsoidal constant energy surface for Si, near the 6 conduction band minima along the *X* directions. (From Chelikowsky and Cohen, Phys. Rev. B14, 556, 1976).

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

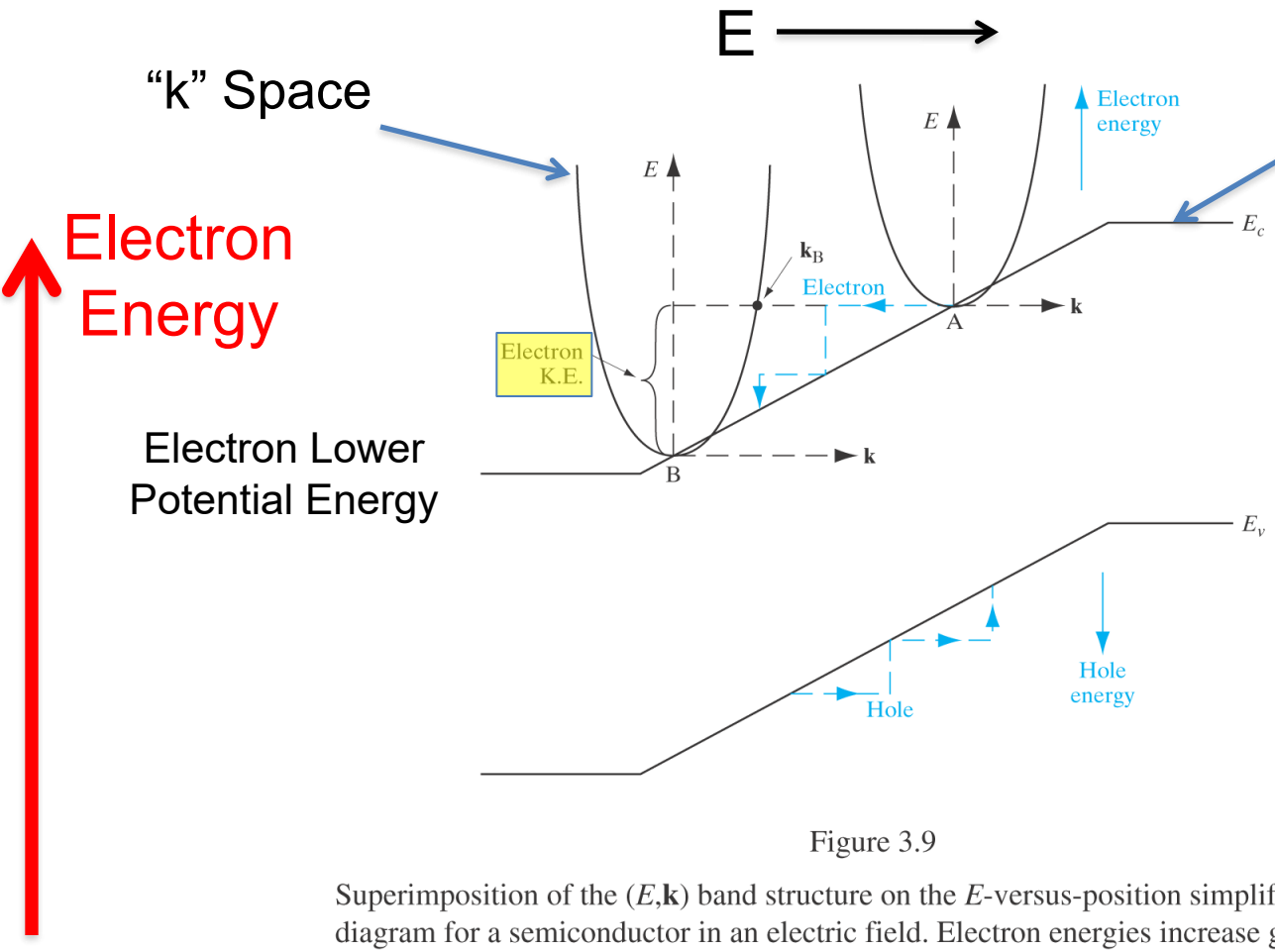
Effective Mass in Si, Ge, and GaAs

- In GaAs
  - Lower effective mass electron in Γ Valley
  - Higher effective mass electron in L Valley
- Applied Electrical Field: Effect on Electron
  - Electron gains energy, increases velocity in Γ Valley
  - Electron collides with lattice, emits a phonon, and transfers to the L valley to become a higher effective mass electron



	Ge	Si	GaAs
m_n^*(or m_e^*)	$0.55m_o$	$1.1m_o$	$0.067m_o$
m_p^*(or m_h^*)	$0.37m_o$	$0.56m_o$	$0.48m_o$

Inferring Electron Potential & Kinetic Energy From Band Diagrams



- The location of the band edge is an indication of the potential energy
- The location above the band edge is an indication of kinetic energy

Figure 3.9

Superimposition of the (E,**k**) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wave vectors point in opposite directions and these charge carriers move opposite to each other, as shown.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee. ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

What about "Hole" Potential & Kinetic Energy?

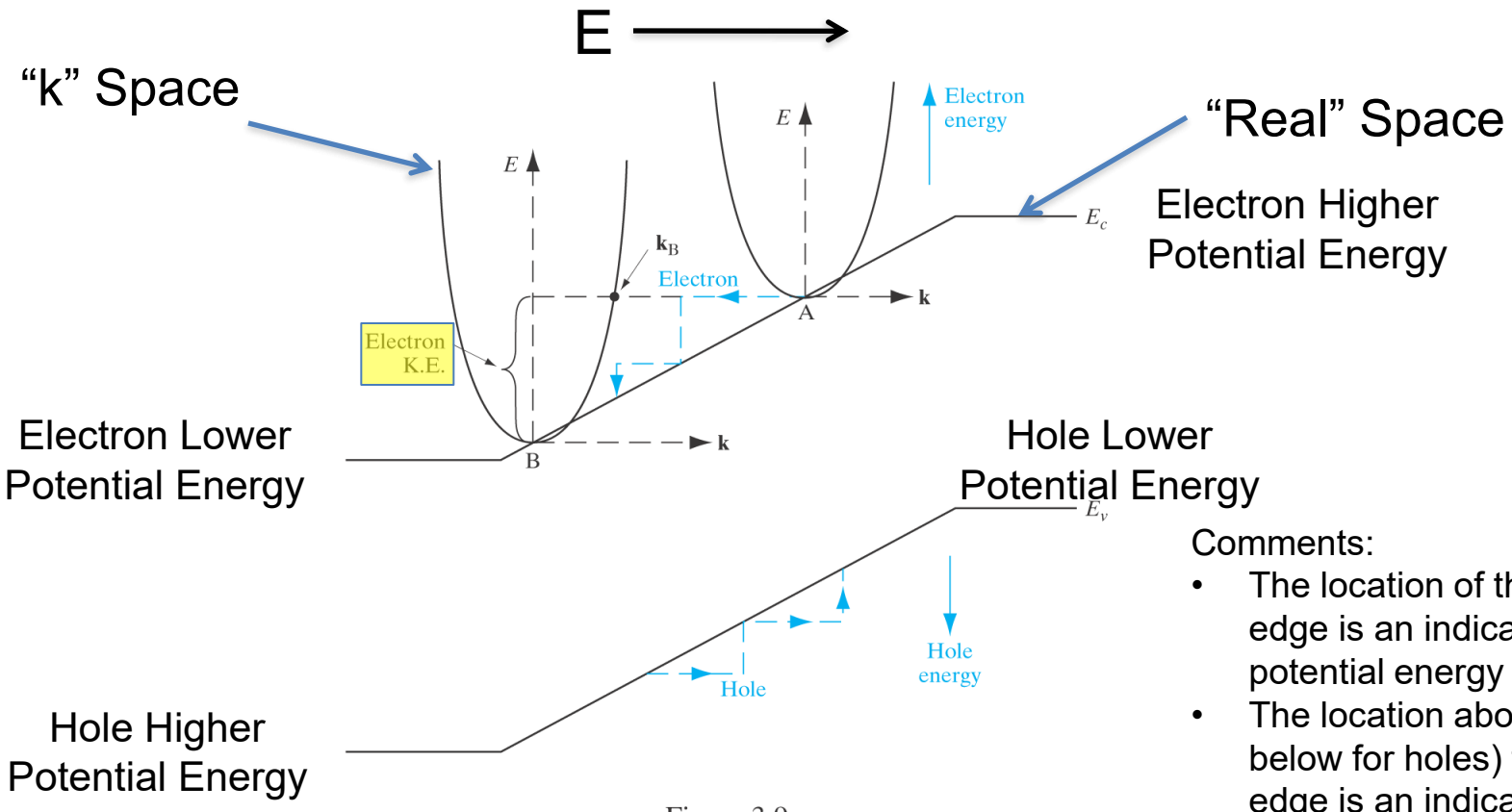


Figure 3.9

Superimposition of the (E,**k**) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wave vectors point in opposite directions and these charge carriers move opposite to each other, as shown.

Comments:

- The location of the band edge is an indication of the potential energy
- The location above (or below for holes) the band edge is an indication of kinetic energy

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee. ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

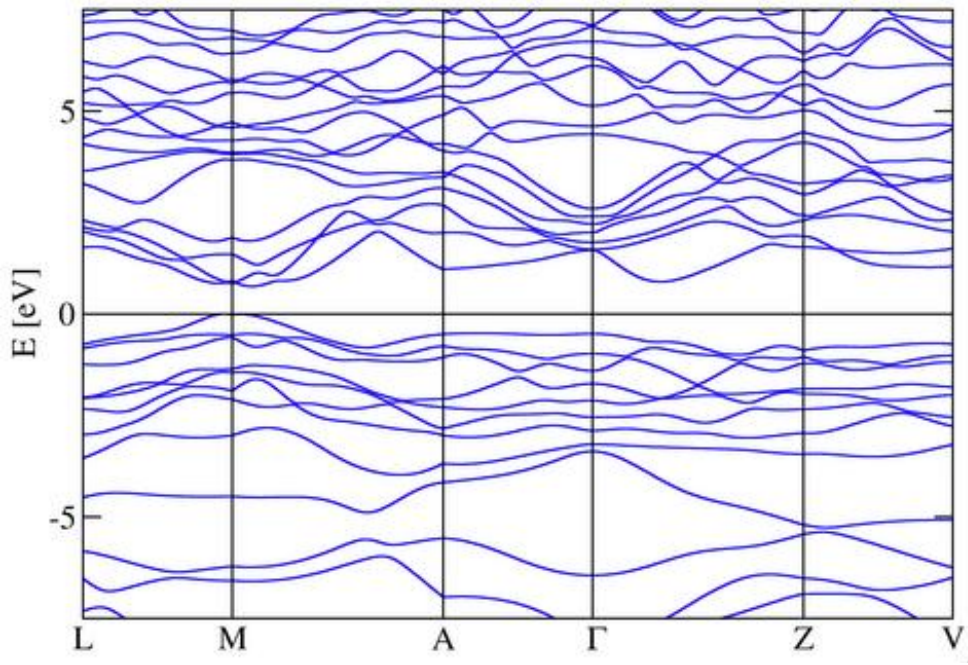
A Few Cautions

- The topic of effective mass is more complicated than can reasonably be presented in an introductory course (treatment as tensor in higher level courses)
- **Make sure you know what effective mass to use and how to use it based upon what you are calculating**
- Calculation of the carrier concentration uses the **density of states effective mass**
  - For Silicon, the density of states effective mass is $(m_n^*)^{3/2} = 6(m_l m_t^2)^{1/2}$
  - This comes from the 6 indirect valleys, and the elliptical shape of those valleys
  - Direct-gap materials such as GaAs have a single minima, so just the effective mass is used
- **Conductivity effective mass** is different than the density of states effective mass for Silicon $$m_n^* = \frac{3}{(\frac{1}{m_l} + \frac{1}{m_t} + \frac{1}{m_t})}$$
- **Cyclotron effective mass** is also different
- There is further complication around the direct-indirect crossover in a material where multiple valleys can be populated

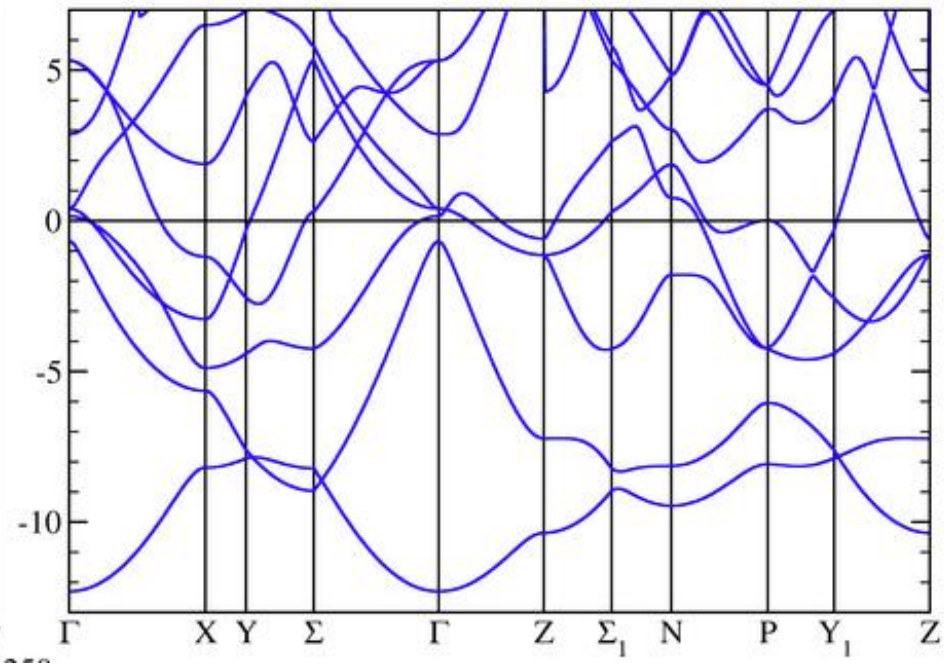
Metals, Semiconductors, and Insulators

Metal and Semiconductor in k-Space

Semiconductor



Metal



- ***Semiconductor/Insulator:*** Maximum valence band energy is lower than minimum conduction band energy
- ***Metal:*** Maximum valence band energy is higher than minimum conduction band energy (band overlap), or partially filled bands

http://www.nature.com/srep/2013/130315/srep01466/full/srep01466.html

Energy Band Picture

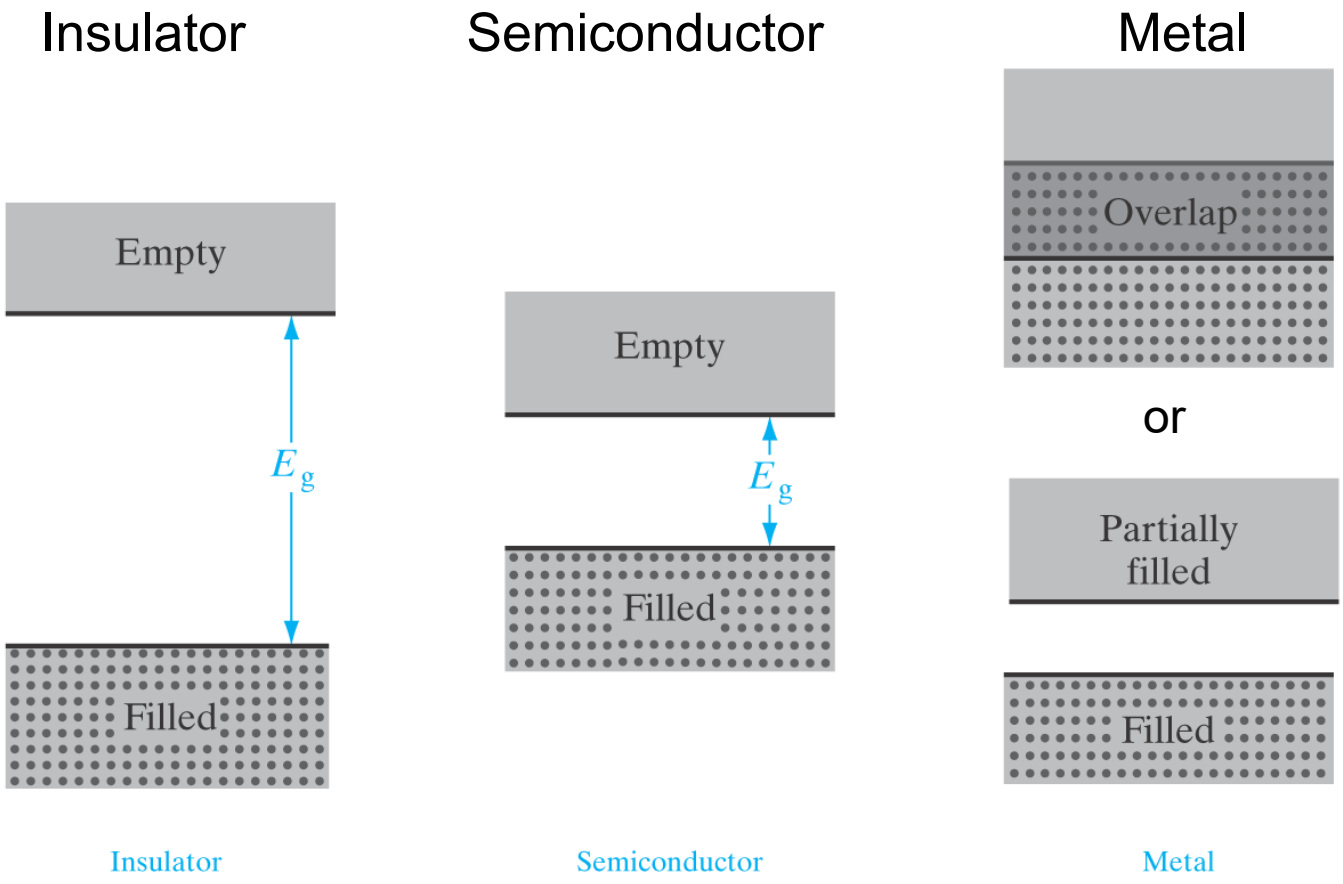


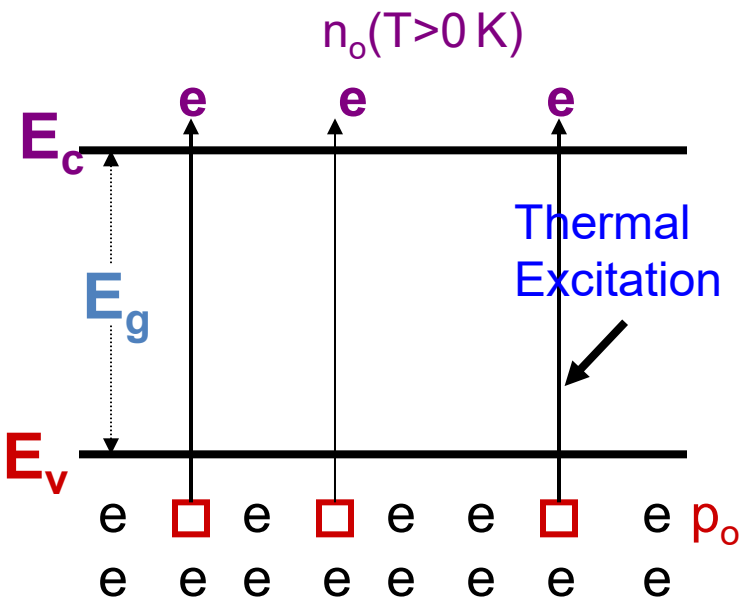
Figure 3.4

Typical band structures at 0 K.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Definitions

Conduction Band Nx4 States



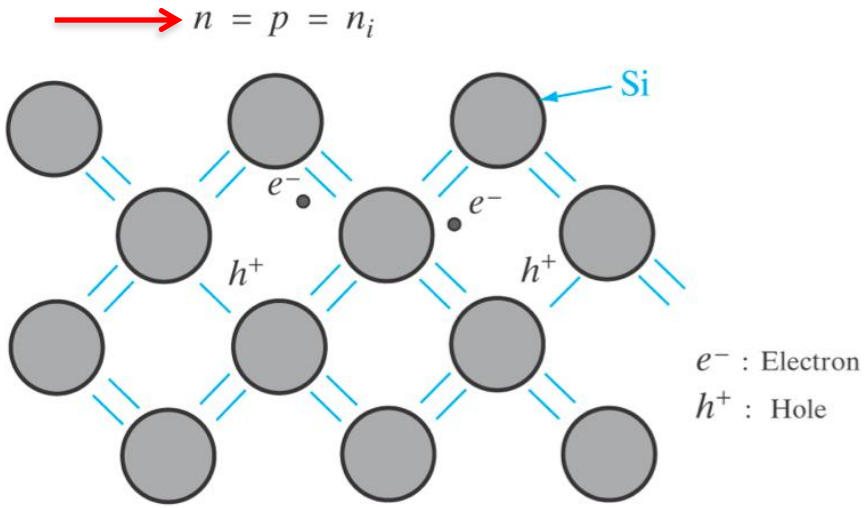
Valence Band: Nx4 electrons occupy Nx4 states at 0 K

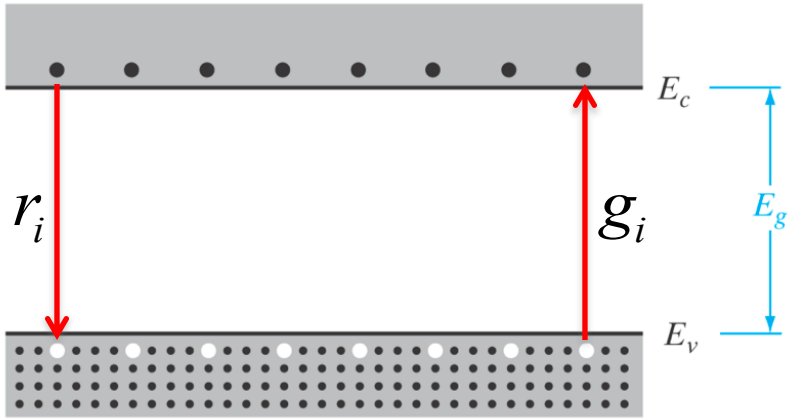
- ***Insulator:*** *no electrons are present in the conduction band at T=0 K and the energy gap is large (Eg > ~10 eV). Virtually no electrons can be thermally excited from the valence band to the conduction band at T=300 K.*
- ***Semiconductor:*** *no electrons are present in conduction band at T=0 K and the energy gap is smaller (~0.1< Eg < ~5 eV). Some electrons are thermally excited from the valence band to the conduction band at T=300 K.*
- ***Metal****: Some electrons are present in the conduction band at T=0 K and the conduction band is either partially filled or overlaps valence band.*

Intrinsic and Extrinsic Material

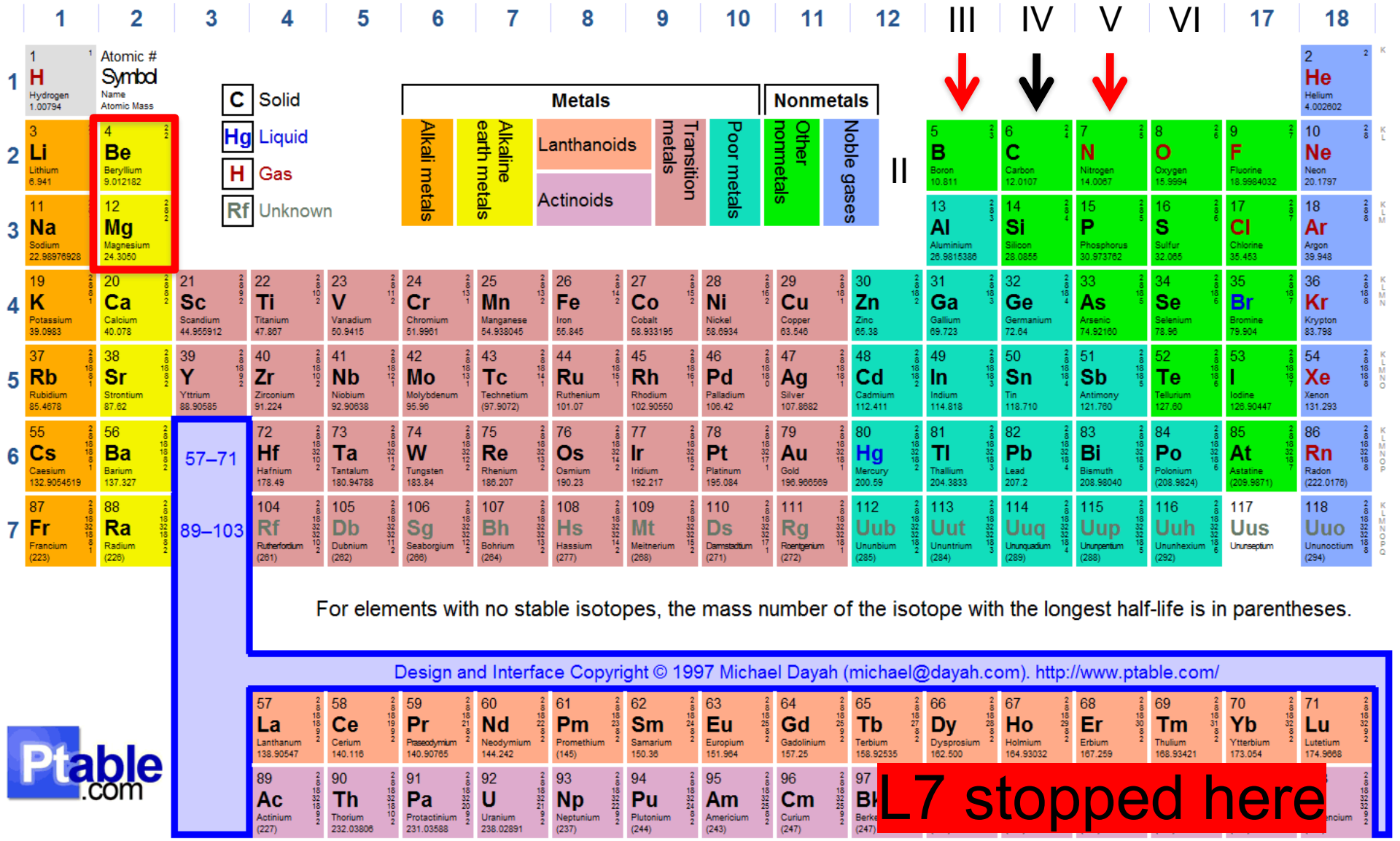
Intrinsic Material

- Intrinsic material: No impurity (no donor or acceptor)
- Electrons and holes present in equal concentrations, called the ***intrinsic concentration, n_i***
- n_i of Si at 300K ~ 1.5×10^{10} cm^{-3}
- The intrinsic concentration n_i increases with increasing temperature (more thermal energy)
- The intrinsic calculation is based upon thermal energy only – no light, current injection, etc.



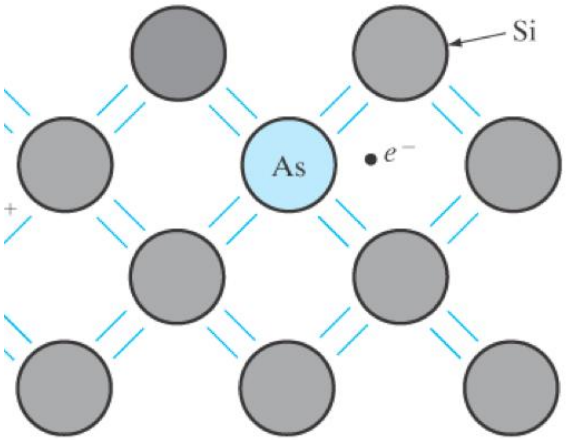


What Happens if We Insert an Atom from Another Column?

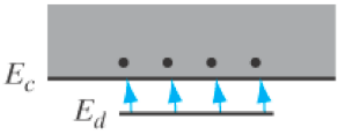


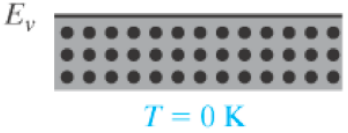
Extrinsic Material

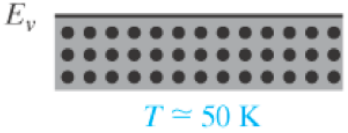
As impurity (donor)



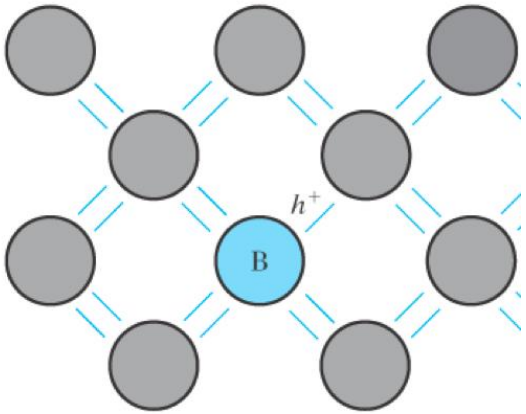






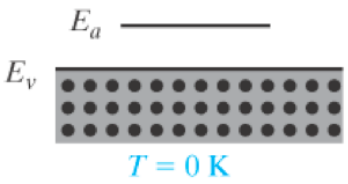


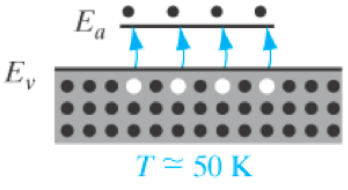
B impurity (acceptor)









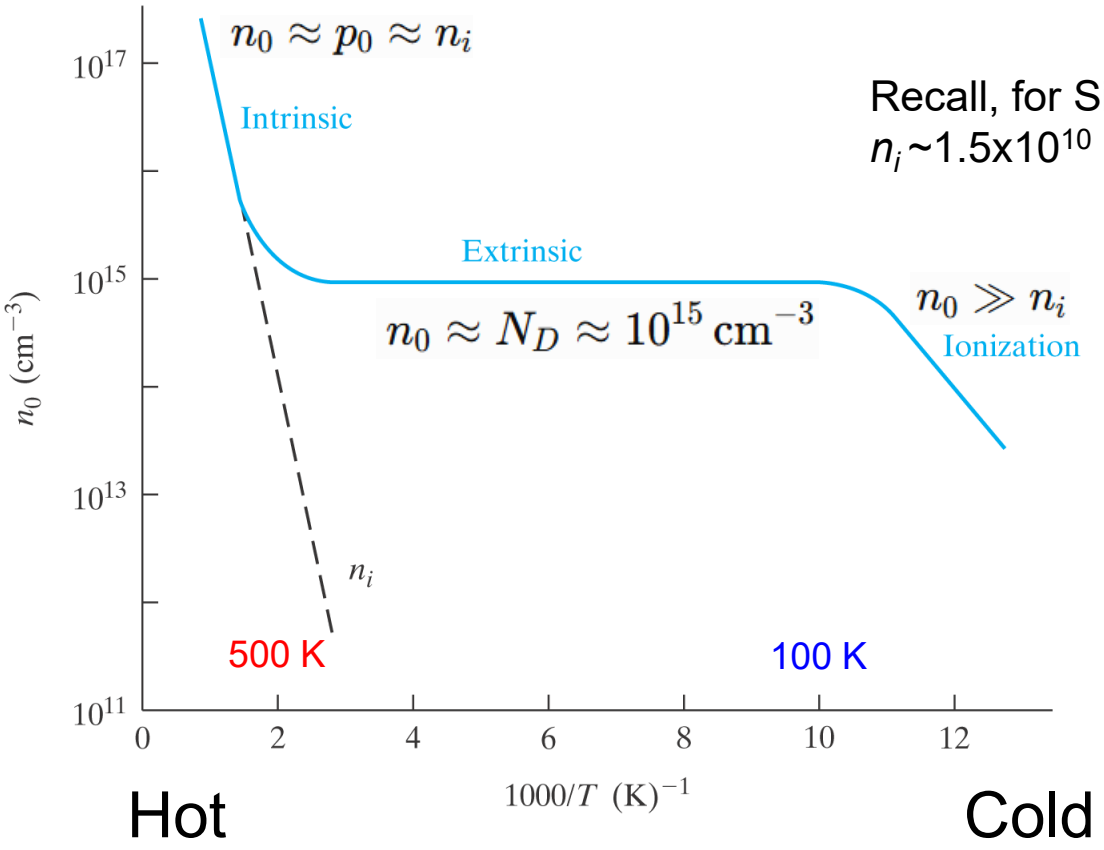


From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Extrinsic Material: Temperature Effects

Carrier concentration in Si doped with 10^{15} donors/cm^3

n_0: Electron concentration (in conduction band)



Recall, for Si, n_i ~1.5x10^{10} cm^{-3} at 300K

Figure 3.18

Carrier concentration vs. inverse temperature for Si doped with 10^{15} donors/cm^3.

From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Important Definitions

- **Intrinsic Material:** For a "perfect" crystal in equilibrium, electron and hole densities are equal
  - The value of this density n_i is called the intrinsic concentration
- **Extrinsic Material:** The crystal is "doped" with
  - **donors**, where the material is n-type and the electron density exceeds the hole density, or
  - **acceptors**, where the material is p-type and the hole density exceeds the electron density
  - An **amphoteric dopant** can act as either a donor or acceptor, depending upon what atom it is replacing in the crystal
  - In extrinsic material, the carrier with the higher concentration is the **majority carrier** and the carrier with lower concentration is the **minority carrier**
    - We will find out later that the product of the minority carrier density and majority carrier density is constant for a given temperature (in equilibrium), and equal to n_i^2
    - In p-type material, holes are the majority carrier, electrons the minority carrier
    - In n-type material, electrons are the majority carrier, holes the minority carrier